



# Reinforcement Learning for High-Frequency Trading

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## Abstract

This study builds upon the memory-augmented context-aware reinforcement learning framework proposed in the MacroHFT paper, optimizing its architecture and conducting performance comparisons for high-frequency trading (HFT). Based on the two-stage training framework of MacroHFT, we further refined the training process of sub-agents and the decision-mixing mechanism of the hyper-agent. First, we introduced a dynamic weight adjustment mechanism to enhance the sub-agents' ability to adapt their strategies to varying market trends and volatility, reducing overfitting and improving responsiveness to rapid market changes. Second, we upgraded the original DQN algorithm to Rainbow DQN and QR-DQN, enhancing the hyper-agent's memory mechanism to improve long-term market pattern recognition and achieve more stable decision-making in extreme market conditions. Third, we optimized the internal neural architecture by incorporating the Multi-Patch Former, a Transformer-based model tailored for financial time series, allowing more effective extraction of local temporal dependencies and global context features from noisy market data.

## Research objectives

The present study investigates the following objectives:

- **Objective 1:** Enhance the adaptability and robustness of the MacroHFT framework through architectural and algorithmic improvements.
- **Objective 2:** Leverage advanced sequence modeling techniques, such as the Multi-Patch Former, to better capture patterns in high-frequency financial data.

## Overall Framework

The overall framework of our proposed method consists of two main phases:

- (1) Sub-agent training with a conditional adapter based on market decomposition (trend and volatility)
- (2) Hyper-agent training with memory augmentation for meta-policy learning.
- (3) Augmented Memory mechanism

- **Add:** New experiences  $(s, a, v)$  are added to  $M$  using a FIFO update rule.
- **Lookup:** The top- $m$  most similar entries are retrieved via L2 distance for weighted aggregation, enabling the hyper-agent to generalize across past high-value decisions.

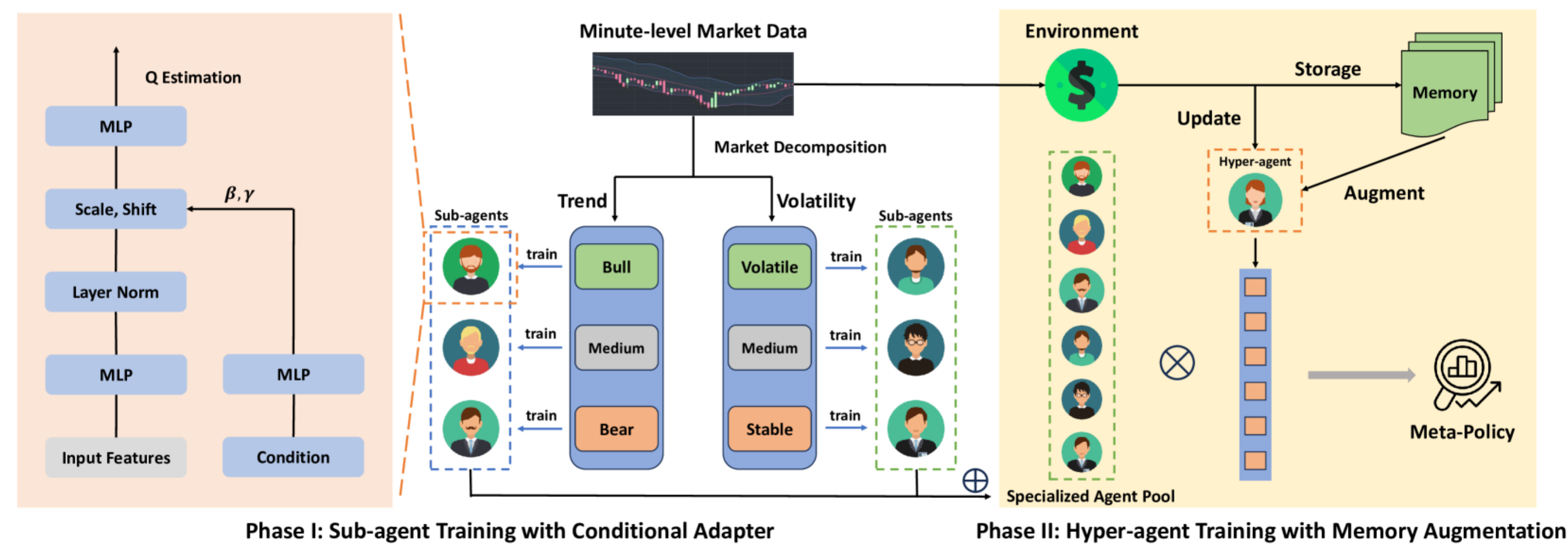


Figure 1. The overall framework: sub-agent training and hyper-agent training with memory augmentation.

## Algorithm exploration

In this study, the primary architecture is based on a hierarchical reinforcement learning framework. The lower-level agent utilizes **Double DQN (DDQN)**, while the high-level agent adopts **Dueling DQN** to manage meta-level decision-making.

To enhance performance and robustness, we further experimented with the following algorithms:

- **C51 (Categorical DQN):** An advanced distributional variant from the Rainbow DQN family.
- **QR-DQN (Quantile Regression DQN):** A distributional reinforcement learning method that models return distributions via quantile approximation.

## Algorithm exploration (cont.)

- **Dynamic soft-hard hybrid strategy for high-level agent:** This includes analyzing market conditions, assigning adaptive weights, and dynamically combining decision outputs to adjust to varying market regimes.
- **Multi-Patch Former:** A Transformer-based architecture designed for financial time series.

## Experiment Setup

- Cryptocurrency markets: **ETHUSDT**, using minute-level high-frequency trading data.
- OHLC and LOB data are segmented, with **ETHUSDT** in 4320-minute blocks. Each is further divided into six subsets based on trend and volatility.
- A low-pass filter is applied, followed by linear regression to compute the slope. Based on the slope, each segment is classified into bull, stable, or bear markets.
- Average volatility is calculated and categorized into high, medium, and low. The validation set is labeled using volatility quantiles derived from the training set.

## Results and discussion

### Model estimates

To briefly assess model performance, we compute **cumulative return** and **maximum drawdown**, providing a preliminary view of profitability and risk exposure.

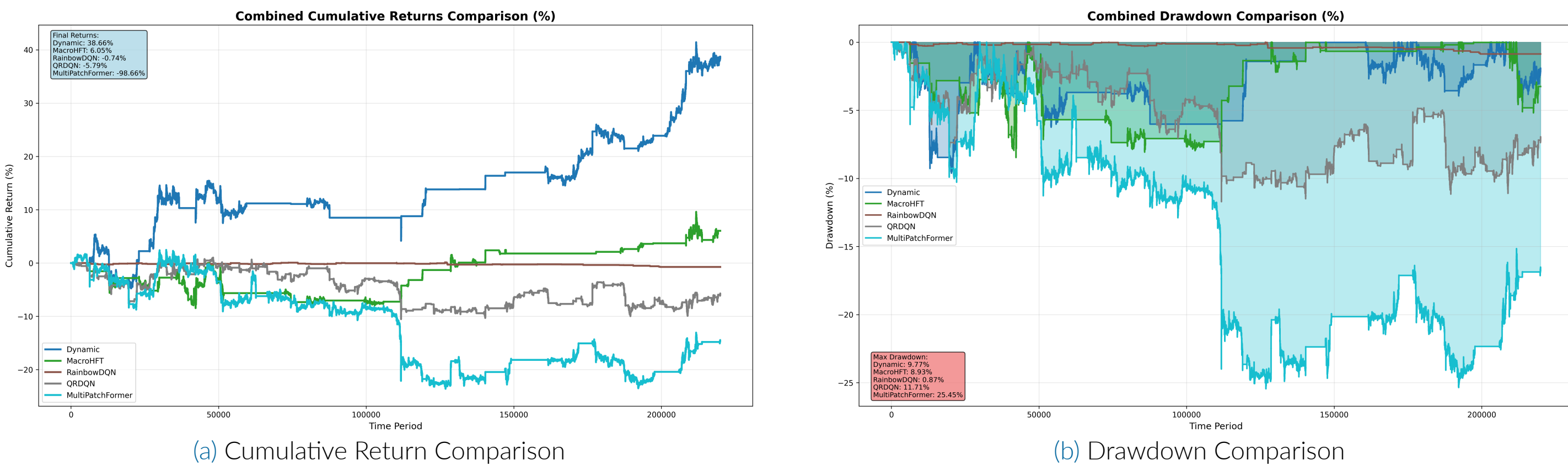


Figure 2. Performance comparison of different strategies.

To evaluate model performance across different trading strategies, we employed a combination of profitability metrics, risk-adjusted measures, and behavioral statistics.

Table 1. Comprehensive performance metrics

| Metric                | MacroHFT | QR-DQN    | Rainbow DQN | Dynamic Hybrid | Multi-Patch Former |
|-----------------------|----------|-----------|-------------|----------------|--------------------|
| Win Rate              | 0.6333   | 0.5651    | 0.4366      | 0.6667         | 0.4872             |
| Avg Profit/Trade      | 0.811    | -0.0864   | -0.0394     | 7.7765         | -1.6061            |
| Avg Profit % / Trade  | 0.002    | -0.000215 | -0.000104   | 0.021479       | -0.003724          |
| Avg Win               | 4.619    | 1.6359    | 0.2069      | 15.5167        | 7.2205             |
| Avg Loss              | -5.7667  | -2.324    | -0.2303     | -7.7037        | -9.9914            |
| Profit Factor         | 1.3836   | 0.9145    | 0.6963      | 2.0134         | 0.6865             |
| Expectancy            | 0.811    | -0.0864   | -0.0394     | 7.7765         | -1.6061            |
| SQN                   | 0.6816   | -0.494    | -1.180      | 2.1713         | -0.8107            |
| Total Profit          | 24.3387  | -23.2473  | -2.7973     | 139.977        | -425.5736          |
| Total Return          | 0.06     | -0.0579   | -0.0073     | 0.38662        | -0.9866            |
| Annualized Return     | 0.15     | -0.1327   | -0.0175     | 1.1838         | -0.99997           |
| Sharpe Ratio          | 0.867    | -0.8045   | -1.8288     | 3.643          | -1.3811            |
| Sortino Ratio         | 0.468    | -0.4809   | -0.0808     | 2.9678         | -1.3039            |
| Max Drawdown          | 0.089    | 0.1171    | 0.008651    | 0.0977         | 0.254468           |
| Max Drawdown Duration | 82018    | 189902    | 93266       | 93740          | 187671             |
| Calmar Ratio          | 1.685    | -1.1336   | -2.0229     | 12.116         | -3.9296            |
| Total Fees            | 4.1672   | 37.5853   | 9.9994      | 2.3901         | 5.4656             |

## Conclusions

- **Superior Performance:** The proposed Dynamic Hybrid method achieved 38.66% cumulative return and 3.643 Sharpe ratio, significantly outperforming traditional MacroHFT (6.05%) and other baseline methods on ETHUSDT market.
- **Mixed Results with Insights:** While some individual algorithms (Rainbow DQN, QR-DQN, Multi-Patch Former) underperformed compared to the original MacroHFT, they provided valuable insights for developing the superior Dynamic Hybrid approach through strategic integration.